

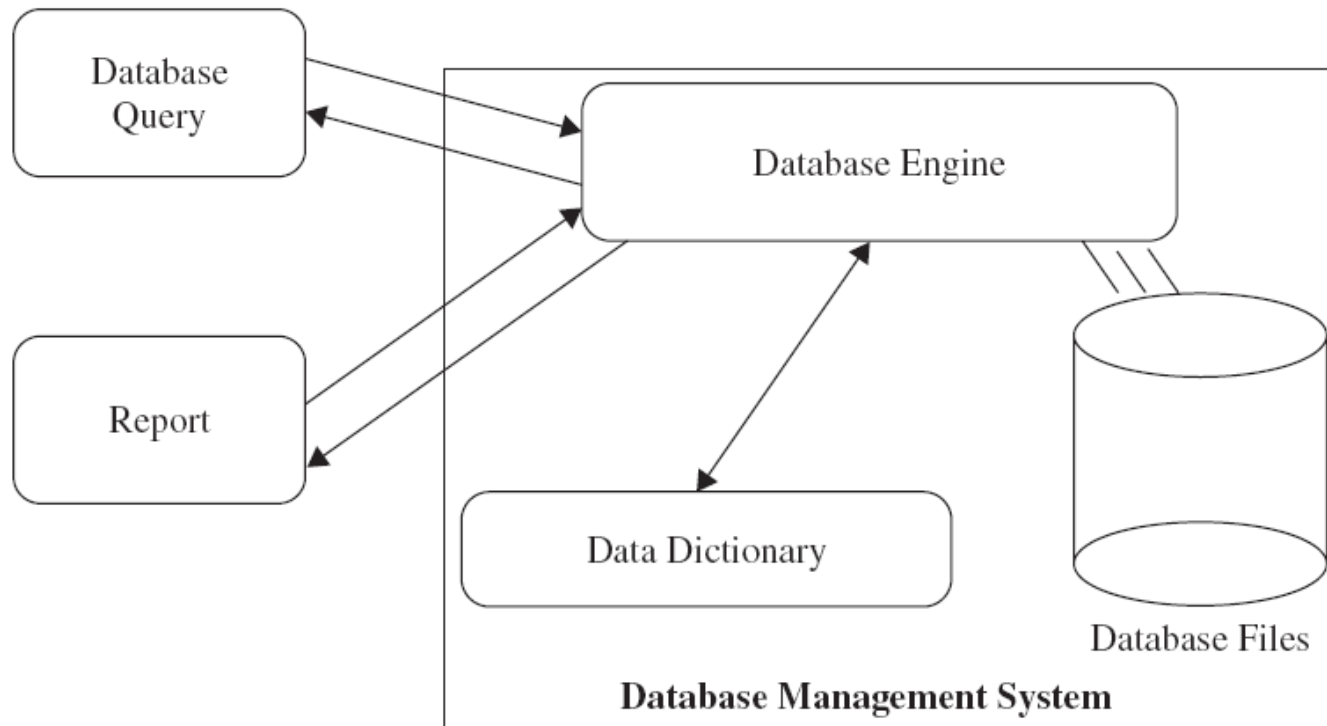
Introduction to PostgreSQL

Outline

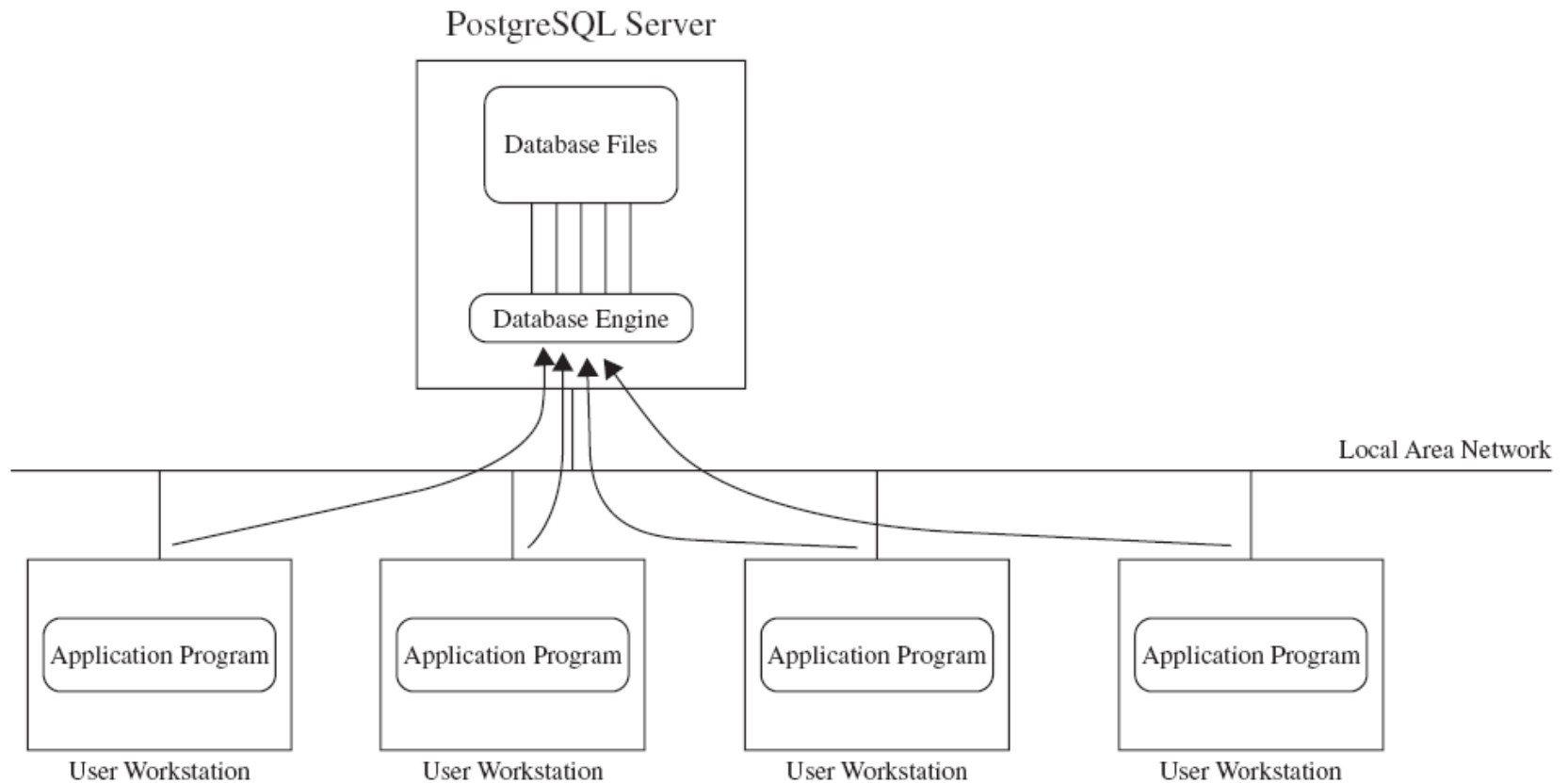
1. Database environments
2. Comparing PostgreSQL
3. Installing PostgreSQL on Windows
4. The PostgreSQL Files and Programs

1. Database environments

A Simple Database engine



A multiuser PostgreSQL environment



PostgreSQL

- **Scalable.** Vertical scalability is a hallmark of PostgreSQL. Considering that almost any custom software solution tends to grow, resulting in database extension, this particular option certainly supports business growth and development.
- **Support for custom data types.** PostgreSQL natively supports a large number of data types by default, such as JSON, XML, H-Store, and others. PostgreSQL takes advantage of it, being one of the few relational databases with strong support for NoSQL features. Additionally, it allows users to define their own data types.
- **Easily-integrated third-party tools.**
- **Free, Open-source and community-driven.** Postgres is completely open-source and supported by its community, which strengthens it as a complete ecosystem. Additionally, developers can always expect free and prompt community assistance

3. Installing PostgreSQL on Windows

Download

<http://www.postgresql.org/download/windows>

➔ **Download the installer** certified by EDB for all supported PostgreSQL versions

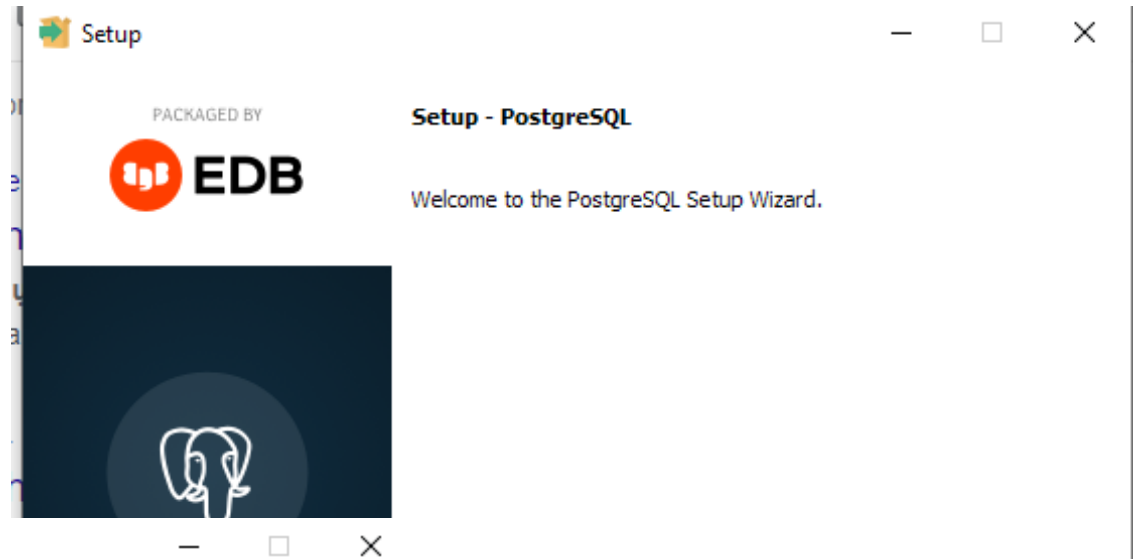
(Latest version 16)

- The graphical installer for PostgreSQL includes
 - the ***PostgreSQL server***
 - ***pgAdmin IV***: a graphical tool for managing and developing your databases
 - and ***StackBuilder***: a package manager that can be used to download and install additional PostgreSQL applications and drivers (ignore to install at the moment)

Install and test

- Install
- Check server service
- Connect to the server from pgAdmin4 / psql

Install



Setup

Installation Directory



Please specify the directory where PostgreSQL will be installed.

Installation Directory



< Back

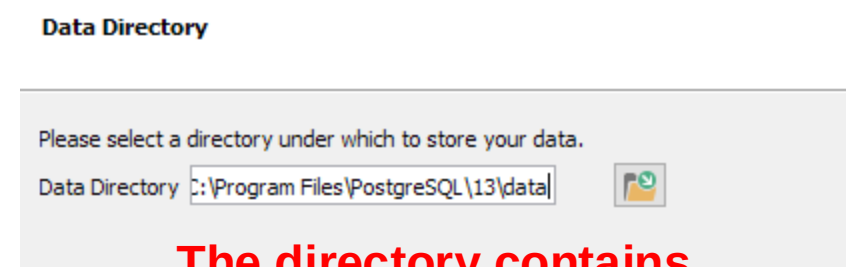
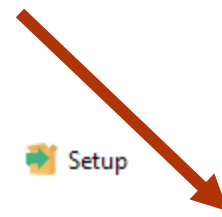
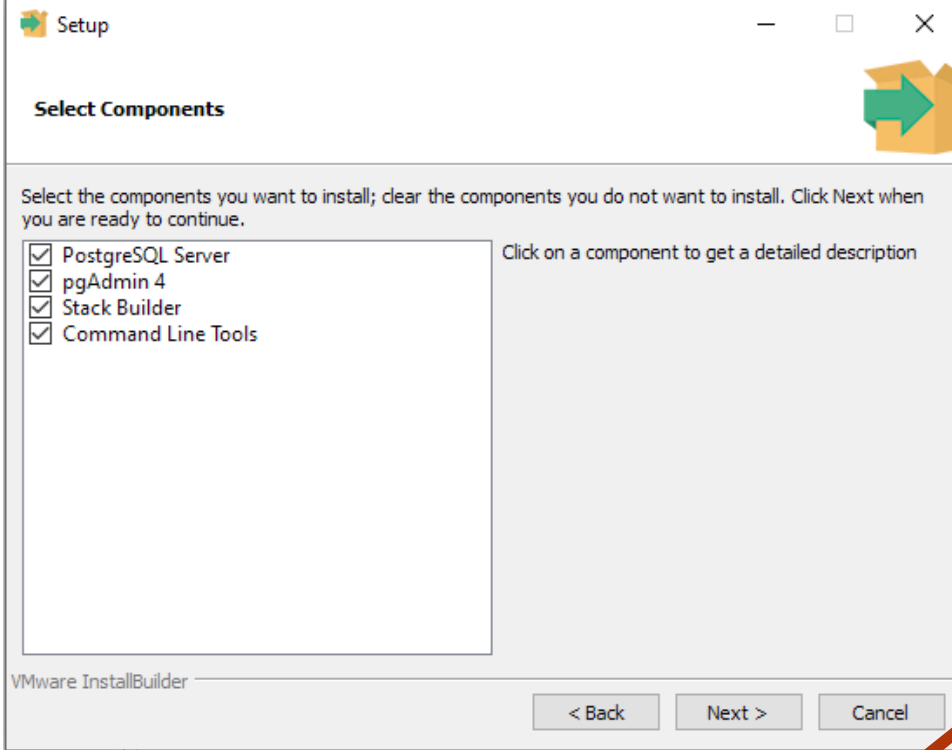
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Cancel

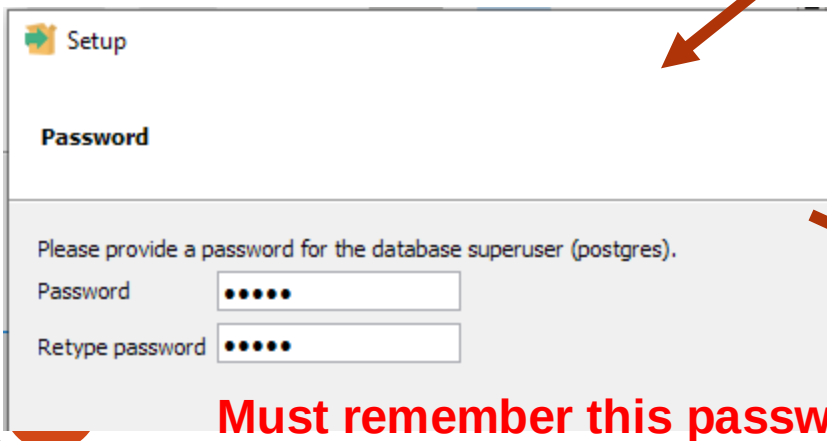
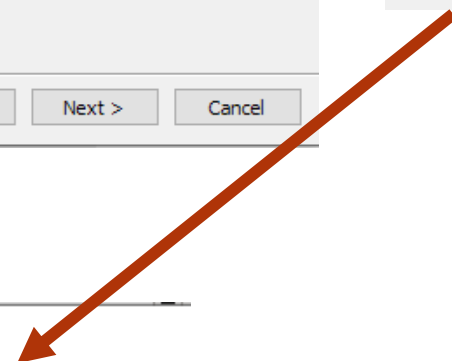
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Next >

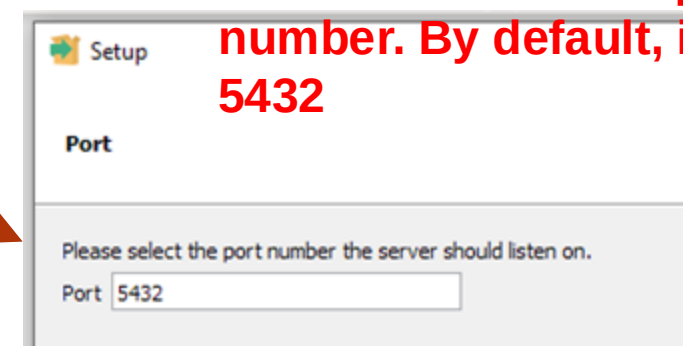
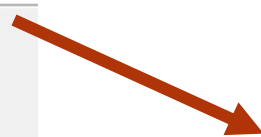
Cancel



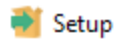
The directory contains all data files



Must remember this password



Must remember the port number. By default, it is 5432

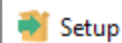


Setup

Advanced Options

Select the locale to be used by the new database cluster.

Locale [Default locale] ▼

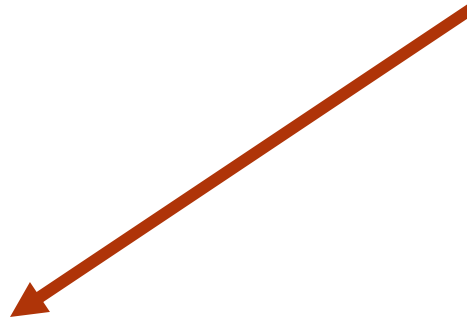
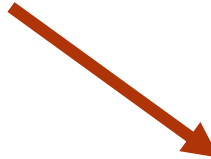


Setup

Pre Installation Summary

The following settings will be used for the installation::

Installation Directory: C:\Program Files\PostgreSQL\13
Server Installation Directory: C:\Program Files\PostgreSQL\13
Data Directory: C:\Program Files\PostgreSQL\13\data
Database Port: 5432
Database Superuser: postgres
Operating System Account: NT AUTHORITY\NetworkService
Database Service: postgresql-x64-13
Command Line Tools Installation Directory: C:\Program Files\PostgreSQL\13
pgAdmin4 Installation Directory: C:\Program Files\PostgreSQL\13\pgAdmin 4
Stack Builder Installation Directory: C:\Program Files\PostgreSQL\13



Setup

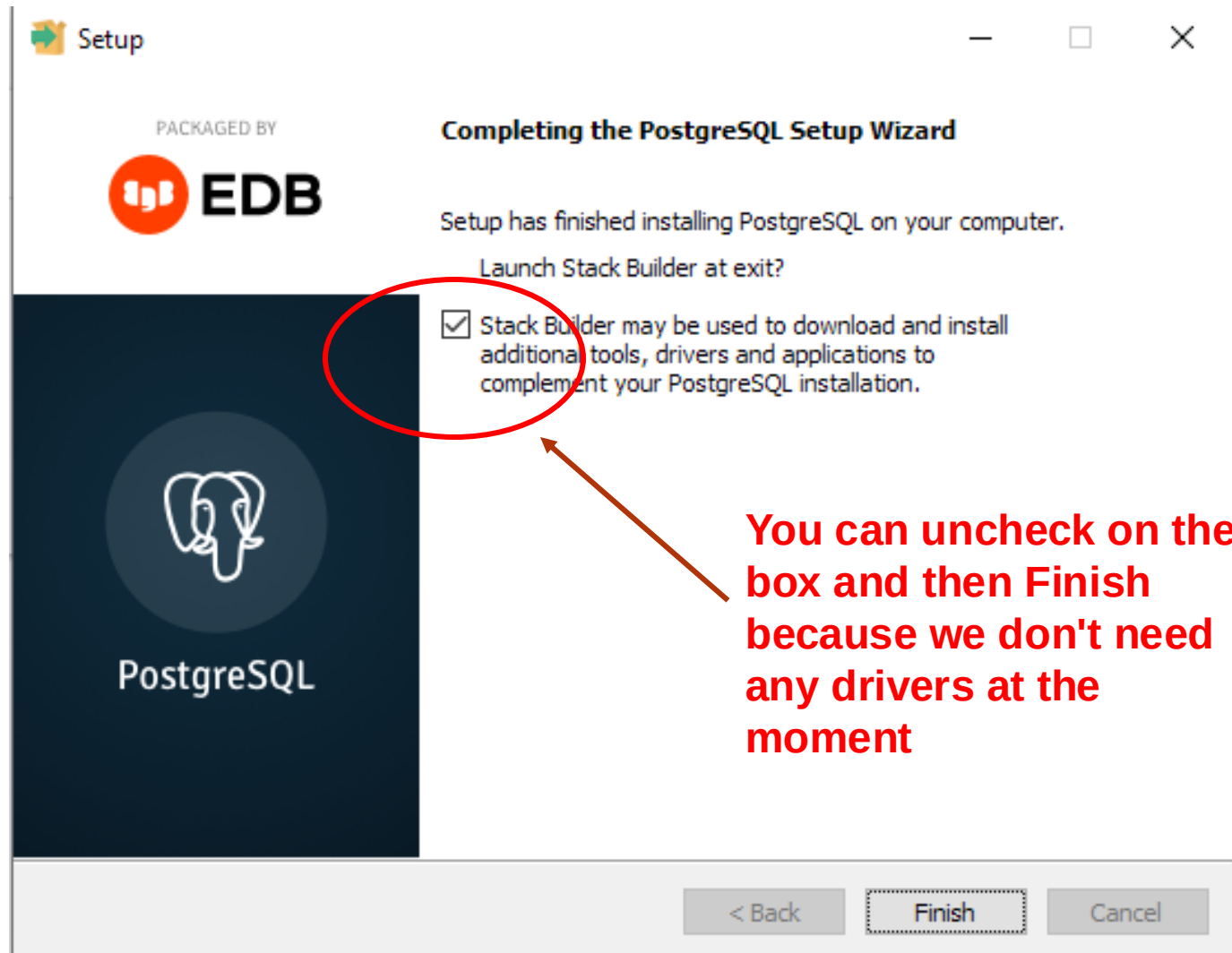
Installing

Please wait while Setup installs PostgreSQL on your computer.

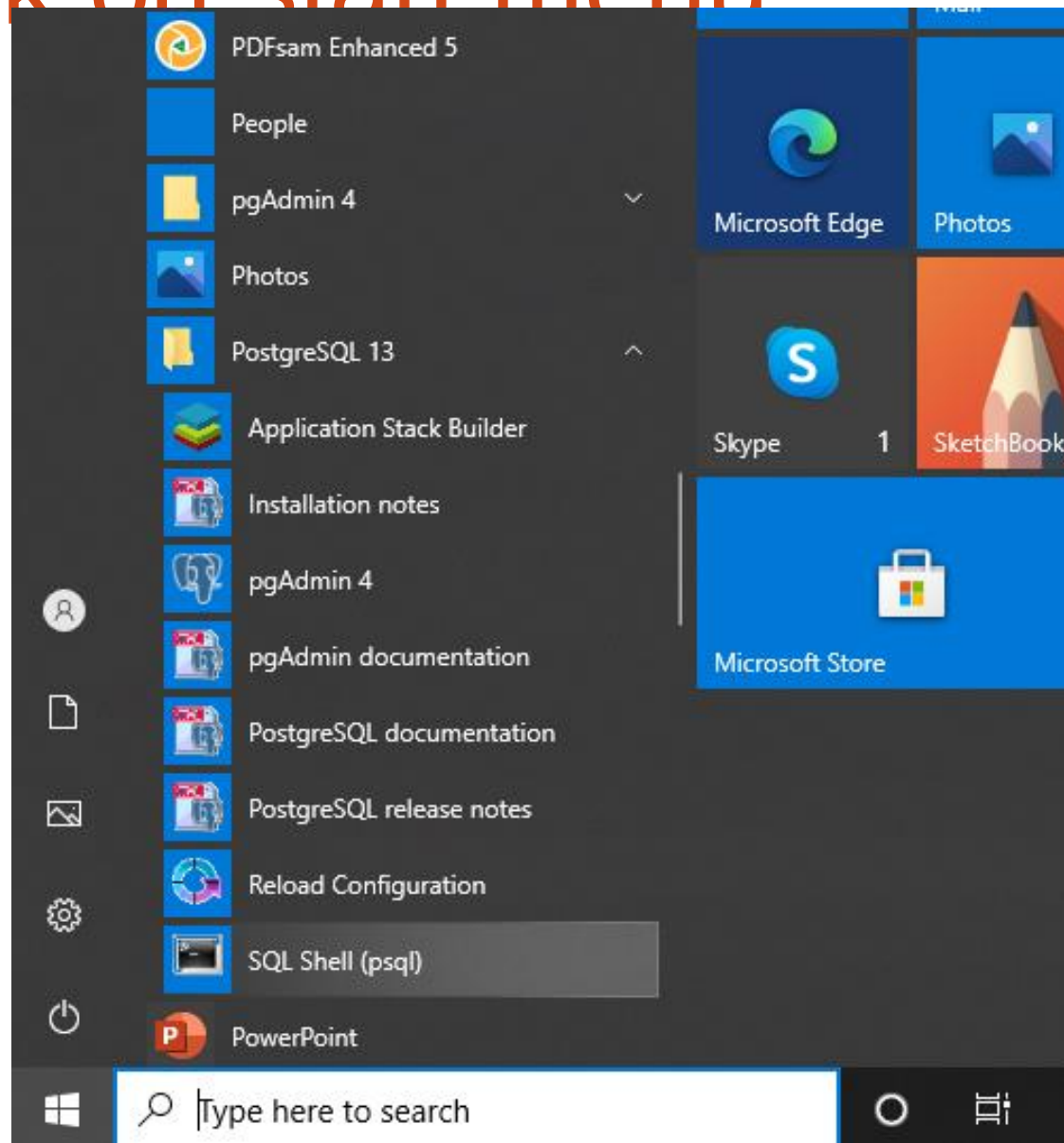
Installing

Unpacking C:\Program [...]\13\doc\postgresql\html\ecpg-sql-prepare.html

Install



Check on start menu

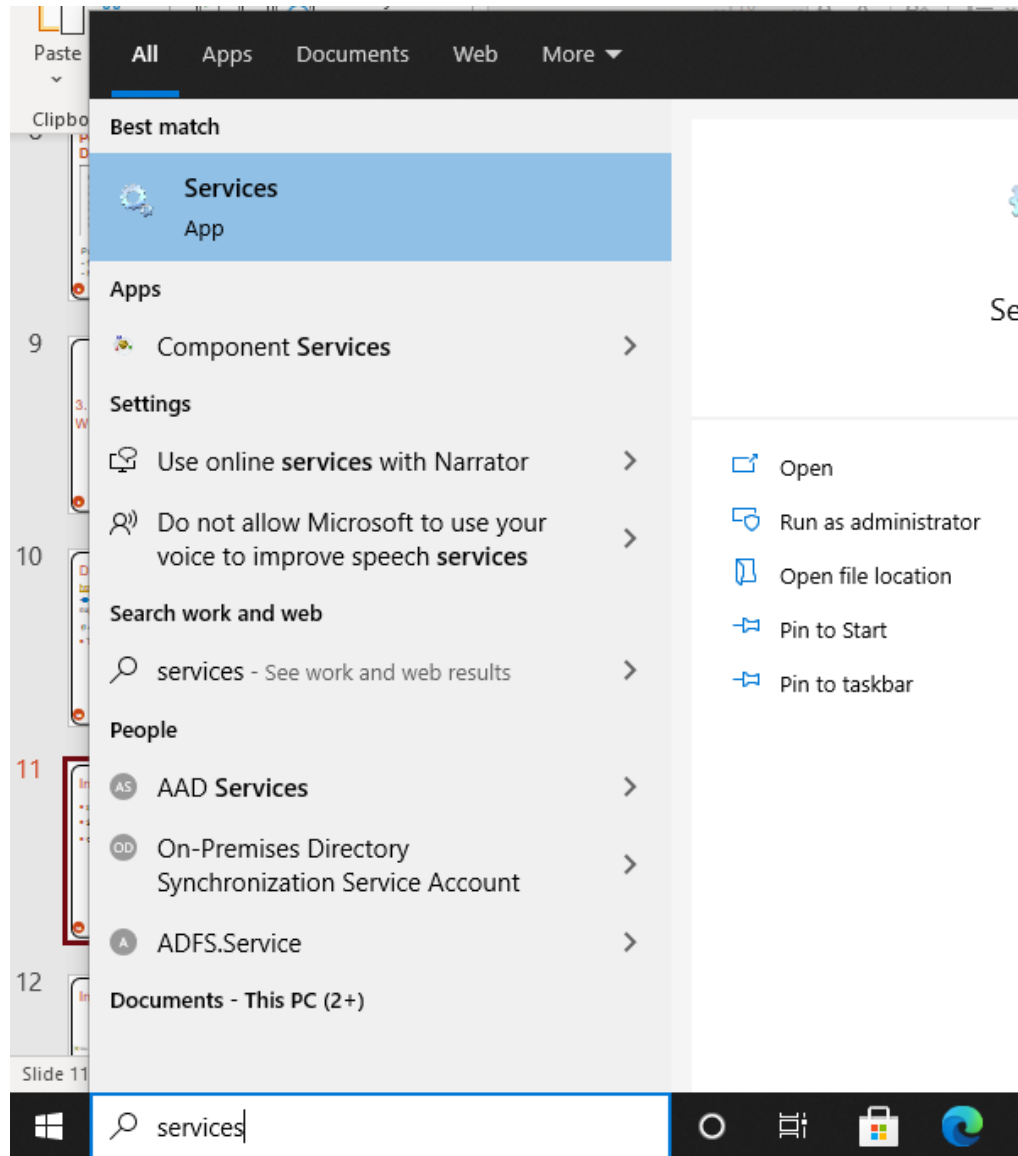


Notes: Uninstall postgresSQL

- Uninstall
- Remove data directory

C:\Program Files\PostgreSQL

Server services



Server services

The server is running

The screenshot shows the Windows Services console. The 'Services (Local)' window is open, displaying a list of services. The 'postgresql-x64-13' service is selected and highlighted in blue. The service's status is 'Running', its startup type is 'Automatic', and it is logged on as 'Network Service'. A red arrow points from the text 'The server is running' to the 'Running' status. Another red arrow points from the text 'You can stop/start/restart the server here' to the 'Stop', 'Pause', and 'Restart' links in the left-hand pane.

Services (Local)

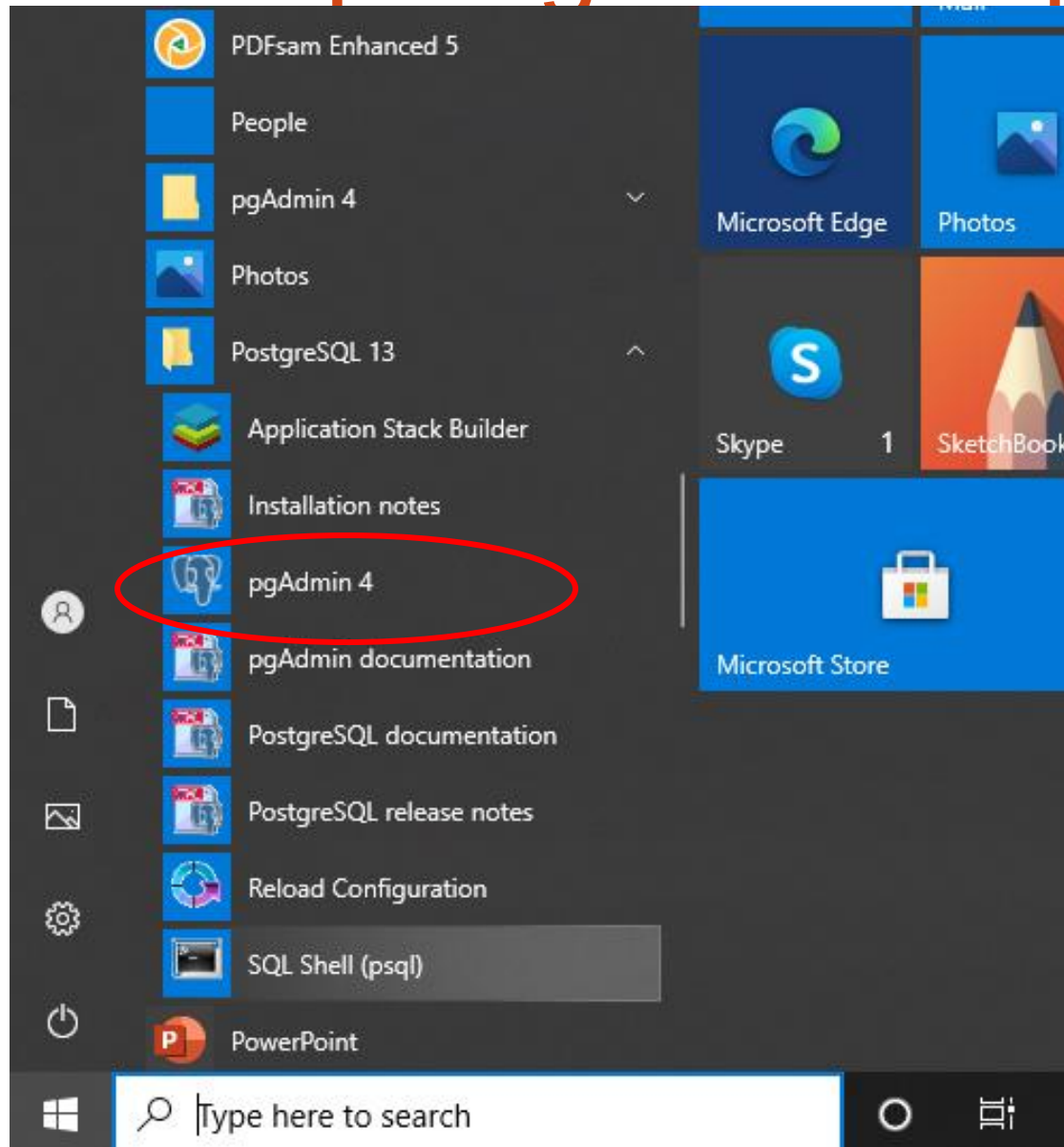
Stop the service
Pause the service
Restart the service

Description:
Provides relational database storage.

Name	Description	Status	Startup Type	Log On As
Parental Controls	Enforces parental controls for child accounts in Windows. If this service i...		Manual	Local System
Payments and NFC/SE Manager	Manages payments and Near Field Communication (NFC) based secure ...		Manual (Trig...	Local Service
PDFsam Enhanced 5		Running	Manual	Local System
PDFsam Enhanced 5 Creator		Running	Automatic	Local System
PDFsam Enhanced 5 Update Service		Running	Automatic	Local System
Peer Name Resolution Protocol	Enables serverless peer name resolution over the Internet using the Peer ...		Manual	Local Service
Peer Networking Grouping	Enables multi-party communication using Peer-to-Peer Grouping. If dis...		Manual	Local Service
Peer Networking Identity Manager	Provides identity services for the Peer Name Resolution Protocol (PNRP)...		Manual	Local Service
Performance Counter DLL Host	Enables remote users and 64-bit processes to query performance counte...		Manual	Local Service
Performance Logs & Alerts	Performance Logs and Alerts Collects performance data from local or re...		Manual	Local Service
Phone Service	Manages the telephony state on the device		Manual (Trig...	Local Service
Plug and Play	Enables a computer to recognize and adapt to hardware changes with lit...	Running	Manual	Local System
Pml Driver HPZ12			Automatic	Local Service
PNRP Machine Name Publication Service	This service publishes a machine name using the Peer Name Resolution ...		Manual	Local Service
Portable Device Enumerator Service	Enforces group policy for removable mass-storage devices. Enables appl...		Manual (Trig...	Local System
postgresql-x64-13	Provides relational database storage.	Running	Automatic	Network Service
Power	Manages power policy and power policy notification delivery.	Running	Automatic	Local System
Print Spooler	This service spools print jobs and handles interaction with the printer. If ...	Running	Automatic	Local System
Printer Extensions and Notifications	This service opens custom printer dialogue boxes and handles notificati...		Manual	Local System
PrintWorkflow_76d7a2b	Provides support for Print Workflow applications. If you turn off this ser...		Manual	Local System
Problem Reports Control Panel Support	This service provides support for viewing, sending and deletion of syste...		Manual	Local System
Program Compatibility Assistant Service	This service provides support for the Program Compatibility Assistant (P...	Running	Manual	Local System
Quality Windows Audio Video Experience	Quality Windows Audio Video Experience (qWave) is a networking platf...		Manual	Local Service
Radio Management Service	Radio Management and Airplane Mode Service	Running	Manual	Local Service
Realtek Audio Service	For cooperation with Realtek audio driver.	Running	Automatic	Local System
Realtek Bluetooth Device Manager Service		Running	Automatic	Local System
Recommended Troubleshooting Service	Enables automatic mitigation for known problems by applying recomm...		Manual	Local System
Remote Access Auto Connection Manager	Creates a connection to a remote network whenever a program referenc...		Manual	Local System
Remote Access Connection Manager	Manages dial-up and individual private network (VPN) connections from th...	Running	Automatic	Local System

You can stop/start/restart the server here

Connect to postgres from pgAdmin 4



Connect to postgres from pgAdmin 4

The image shows the pgAdmin 4 web interface in a browser window. The address bar shows the URL `127.0.0.1:64092/browser/`. The interface has a top navigation bar with 'pgAdmin' and menus for 'File', 'Object', 'Tools', and 'Help'. Below this is a 'Browser' pane on the left showing a tree of servers under 'Servers (4)'. The servers listed are 'PostgreSQL 13', 'barney5@server', 'barney@postgres10', and 'postgres@PostgreSQL 10'. A red arrow points to the 'PostgreSQL 13' entry, and a red text label 'Right click here' is placed next to it. A context menu is open over 'PostgreSQL 13', showing options: 'Create', 'Refresh...', 'Connect Server' (highlighted), 'Remove Server', and 'Properties...'. A red arrow points to the 'Connect Server' option. The right pane shows the 'Properties' tab for the selected server, with a 'General' section visible.

Browser

Servers (4)

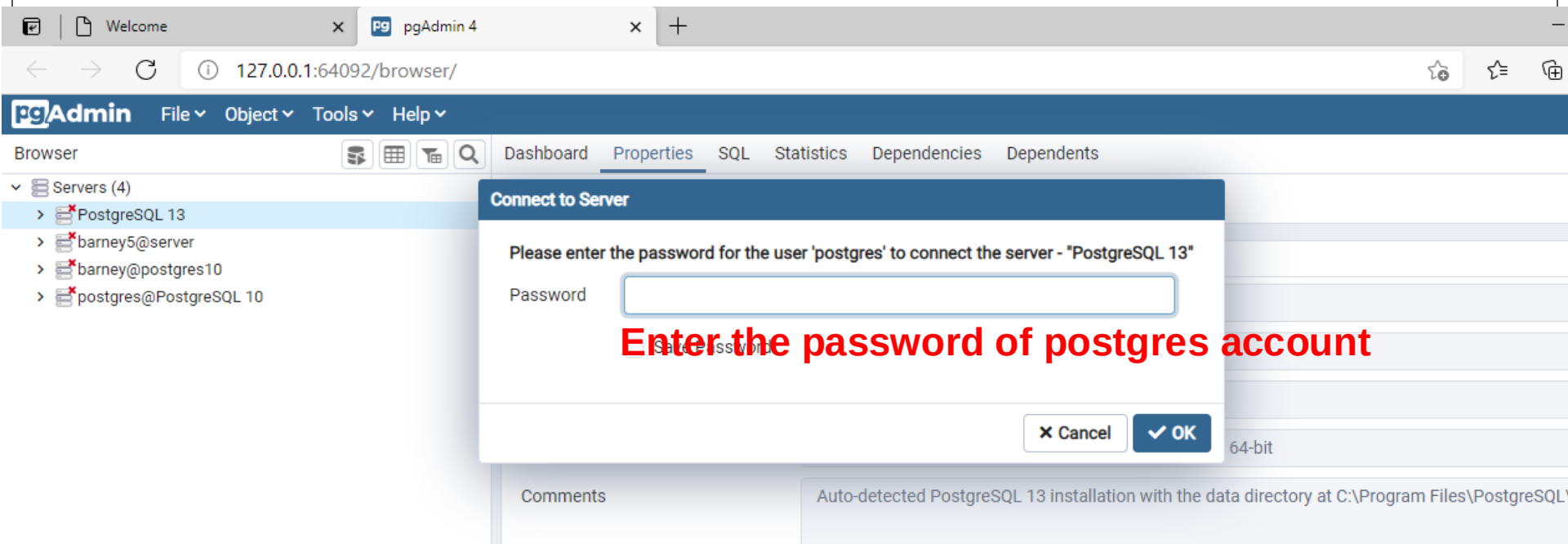
- PostgreSQL 13
- barney5@server
- barney@postgres10
- postgres@PostgreSQL 10

Right click here

Connect Server

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Connect to postgres from pgAdmin 4

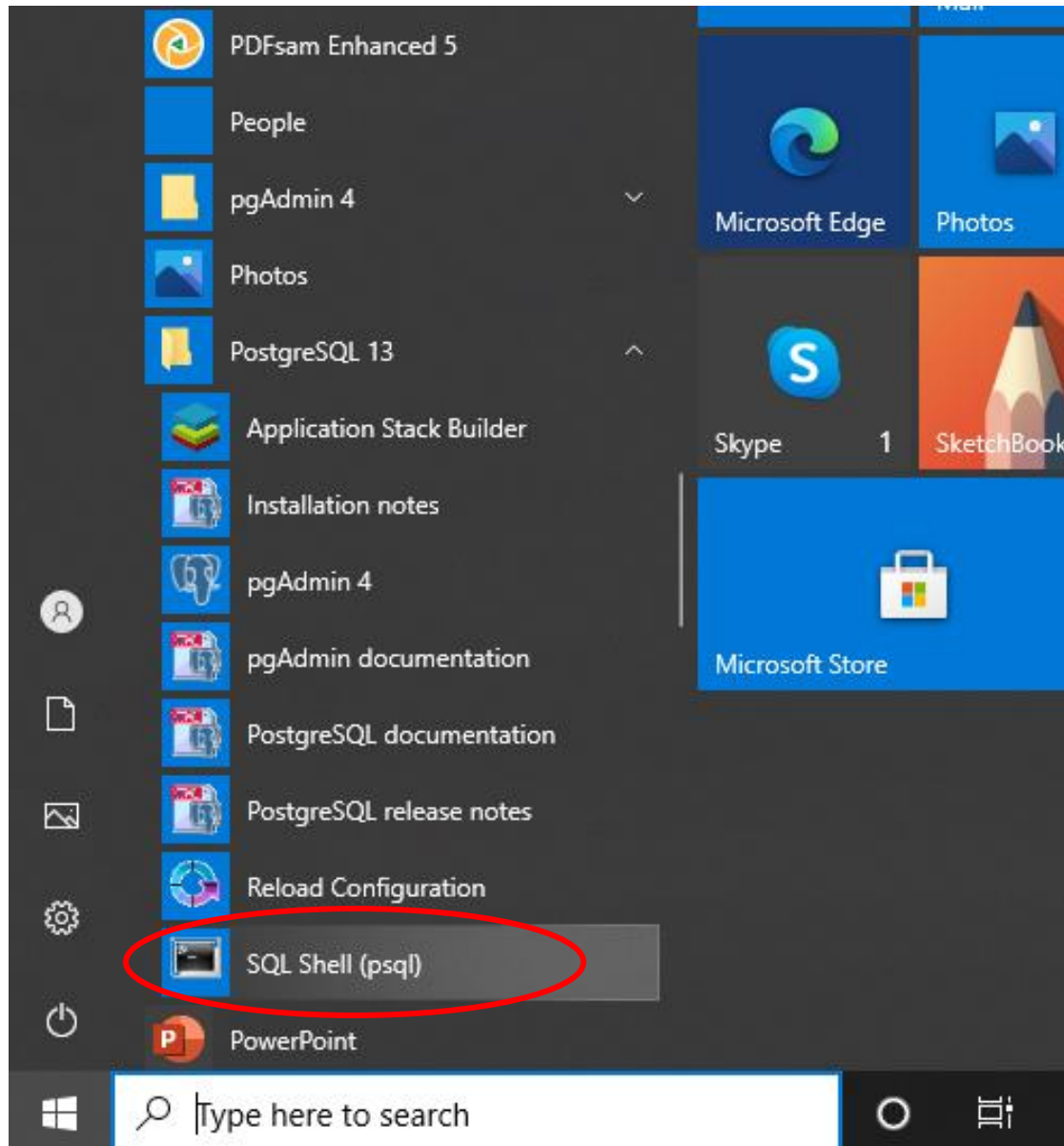


Connect to postgres from pgAdmin 4

The screenshot shows the pgAdmin 4 web interface in a browser. The address bar displays `127.0.0.1:64092/browser/`. The interface has a dark blue header with the pgAdmin logo and navigation menus: File, Object, Tools, and Help. Below the header is a 'Browser' section with a tree view on the left and a 'Properties' panel on the right. The tree view shows a hierarchy: Servers (4) > PostgreSQL 13 > Databases, Login/Group Roles, Tablespaces, barney5@server, barney@postgres10, and postgres@PostgreSQL 10. The 'Properties' panel is currently showing the 'General' tab for the selected 'PostgreSQL 13' server. The 'General' tab displays the following information:

General	
ID	4
Name	PostgreSQL 13
Server type	PostgreSQL
Version	PostgreSQL 13.2, compiled by Visual
Comments	Auto-detected PostgreSQL 13 installa

Connect to postgres using psql



Connect to postgres using psql

```
SQL Shell (psql)
Server [localhost]:
Database [postgres]:
Port [5432]:
Username [postgres]:
Password for user postgres:
psql (13.2)
WARNING: Console code page (850) differs from Windows code page (1252)
        8-bit characters might not work correctly. See psql reference
        page "Notes for Windows users" for details.
Type "help" for help.

postgres=# \l

               List of databases
  Name          | Owner   | Encoding | Collate          | Ctype          | Access privileges
-----+-----+-----+-----+-----+-----
 postgres      | postgres | UTF8      | English_United States.1252 | English_United States.1252 | =c/postgres +
 template0     | postgres | UTF8      | English_United States.1252 | English_United States.1252 | postgres=CTc/postgres
 template1     | postgres | UTF8      | English_United States.1252 | English_United States.1252 | =c/postgres +
                                     postgres=CTc/postgres
(3 rows)

postgres=# \q
```

4. Install PostgreSQL on Ubuntu

Install

Postgresql APT Repository:

(<https://www.postgresql.org/download/linux/ubuntu/>)

Create the file repository configuration:

```
sudo sh -c 'echo "deb http://apt.postgresql.org/pub/repos/apt $(lsb_release -cs)-  
pgdg main" > /etc/apt/sources.list.d/pgdg.list'
```

Import the repository signing key:

```
wget --quiet -O - https://www.postgresql.org/media/keys/ACCC4CF8.asc | sudo  
apt-key add -
```

Update the package lists:

```
sudo apt-get update
```

Install the latest version of PostgreSQL.

If you want a specific version, use 'postgresql-12' or similar instead of 'postgresql':

```
sudo apt-get -y install postgresql
```

Install

- After installing PostgreSQL the following packages are also installed:
(Installed packages: `postgresql`, `postgresql-client`, `postgresql-client-common`, `postgresql-common`, `sysstat` , ...)
- If `pgadmin4` (a graphical administration utility) is not installed, use the `apt-get` command:

```
sudo apt-get install pgadmin4
```

Client installation

- If you only wish to connect to an external PostgreSQL server, install only the PostgreSQL client package:

- Client installation

```
sudo apt-get install postgresql-client
```

- Connect to the server:

```
psql -h server.domain.org database user
```

Example: `psql -h 192.168.100.5 postgres postgres`

```
psql -h localhost postgres postgres
```

Basic Server Setup

- Set the password of the PostgreSQL user (role) called "postgres"
 - Connect as a role with the same name as the local user (i.e. postgres) to the database "postgres"
`sudo -u postgres psql postgres`
 - Set a password for the "postgres" database role:
`\password postgres` and then give your password when prompted
 - Exit the psql prompt: `\q`
- You can try to create the first database with the command
`sudo -u postgres createdb mydb`

Basic Server Setup

- **Allowing local connections**

- By default, local connections are not allowed for the postgres user
- As a super user, open `/etc/postgresql/x.x/main/pg_hba.conf` (Ubuntu) in a text editor:
 - `sudo gedit /etc/postgresql/9.6/main/pg_hba.conf`
- Scroll down to the line that describes local socket connections like this:

<code>local</code>	<code>all</code>	<code>postgres</code>	<code>peer</code>
<code>local</code>	<code>all</code>	<code>all</code>	<code>peer</code>
- Change the "`peer`" method to "`md5`"
- To allow connections using pgAdmin, find the line that describes local loopback connections over IPv6:

<code>host</code>	<code>all</code>	<code>all</code>	<code>:::1/128</code>	<code>ident</code>
-------------------	------------------	------------------	-----------------------	--------------------
- Change the "`ident`" method to "`md5`"
- Save and close the file

Basic Server Setup

- Restart postgresql : `sudo service postgresql restart`
 - To test your connection using **psql**: `psql -U postgres -W`
 - To test your connection using **pgAdmin**, connect to the database at localhost:**5432** using the user name **postgres** and the password supplied

Stop/start/reload PostgreSQL server – on Ubuntu

```
sudo service postgresql  
{start|stop|restart|reload|force-reload|status}
```

Create superuser

- Create a database superuser, same name as login name:

Create a new superuser

```
sudo -u postgres createuser --superuser $USER
```

Set password for the new user

```
sudo -u postgres psql
```

```
postgres=# \password $USER
```


5. The PostgreSQL Files and Programs

PostgreSQL Files – on Windows

- Default: `C:\Program Files\PostgreSQL\x.x`

Directory	Description
bin	The PostgreSQL main programs, utilities, and library files
data	PostgreSQL Data Dictionary, log files, and the transaction log
doc	Documentation on contrib modules, PgOleDb, and psqlODBC
include	C program header files for developing C programs for PostgreSQL (if the Development package was installed)
jdbc	Java JDBC library files for developing Java programs for PostgreSQL (if the JDBC package was installed)
lib	PostgreSQL library files for the executable programs
npgsql	Microsoft .NET library files for developing .NET programs for PostgreSQL (if the npgsql package was installed)
PgAdmin III	The pgAdmin III program documentation
share	Contrib modules and timezone information for PostgreSQL

Database cluster Directory

- Default: `C:\Program Files\PostgreSQL\x.x\data`

Directory	Description
base	Contains a directory for each database
global	Contains system tables for the Data Dictionary
pg_clog	Contains status files on transaction commits
pg_log	Contains PostgreSQL system log files
pg_multixact	Contains multitransaction status information used for row locking
pg_subtrans	Contains subtransaction status information
pg_tblspc	Contains links to database tables
pg_twophase	Contains phase files for the two-phase transaction commit process
pg_xlog	Contains the transaction log files

PostgreSQL Files - on Ubuntu

- **Configuration files:**

`/etc/postgresql/x.x/main/`

- **Data directory:**

`/var/lib/postgresql/x.x/main/`

- **Log files:**

`/var/log/postgres/`

pg_log directory for Administrators

- Each started time, a new log file: *postgres-year-month-day-time.log*
- Each row: a timestamp + the event

```
2006-06-29 20:13:45 FATAL:  database "test" does not exist
2006-06-29 20:24:01 LOG:   transaction ID wrap limit is 2147484148,
                           limited by database "postgres"
2006-06-29 20:25:05 LOG:   autovacuum: processing database "Test"
2006-06-29 20:26:05 LOG:   autovacuum: processing database "template1"
2006-06-29 20:27:05 LOG:   autovacuum: processing database "postgres"
2006-06-29 20:28:18 NOTICE: ALTER TABLE / ADD PRIMARY KEY will create
                           implicit index "ItemID" for table "test"
```

PostgreSQL Log Message Levels

Message Severity	Description
DEBUG	Program information for developers
INFO	Information requested by a database user from a database command
NOTICE	Information that may be useful to the database user regarding a submitted command
WARNING	Information about possible problems in a user session
ERROR	A minor error that caused a user command to abort
LOG	Information of interest for the administrator related to the PostgreSQL system
FATAL	A major error that caused a user session to abort
PANIC	A major error that caused the PostgreSQL system to abort

Configuration Files

- How PostgreSQL behaves is controlled by three separate configuration files
 - *postgresql.conf* (C:\Program Files\PostgreSQL\x.x\data):
 - main parameters for PostgreSQL server (server connection (port, max number of connection), authentication, ...)
 - *pg_hba.conf*:
 - Authentication configuration file for PostgreSQL server allows only specified hosts and defines the server's authentication methods.
 - *pg_ident.conf*
 - configuration file for mapping OS users with PostgreSQL in the authentication method
- Text files:
 - can be changed at any time
 - will not take effect until either the system is restarted or reloaded
 - Each entry in the configuration files is on a separate line
 - #: comment

PostgreSQL Client Applications

Table of Contents

`clusterdb` — cluster a PostgreSQL database
`createdb` — create a new PostgreSQL database
`createuser` — define a new PostgreSQL user account
`dropdb` — remove a PostgreSQL database
`dropuser` — remove a PostgreSQL user account
`ecpg` — embedded SQL C preprocessor
`pg_basebackup` — take a base backup of a PostgreSQL cluster
`pgbench` — run a benchmark test on PostgreSQL
`pg_config` — retrieve information about the installed version of PostgreSQL
`pg_dump` — extract a PostgreSQL database into a script file or other archive file
`pg_dumpall` — extract a PostgreSQL database cluster into a script file
`pg_isready` — check the connection status of a PostgreSQL server
`pg_receivewal` — stream write-ahead logs from a PostgreSQL server
`pg_recvlogical` — control PostgreSQL logical decoding streams
`pg_restore` — restore a PostgreSQL database from an archive file created by `pg_dump`
`pg_verifybackup` — verify the integrity of a base backup of a PostgreSQL cluster
`psql` — PostgreSQL interactive terminal
`reindexdb` — reindex a PostgreSQL database
`vacuumdb` — garbage-collect and analyze a PostgreSQL database

PostgreSQL Client Applications

- **psql:**
 - a command-line interface to the PostgreSQL system
 - Most Unix administrators live and die by simple command-line programs
- **pg_config:** see the current configuration values used to compile and install the PostgreSQL package
- **pg_dump:** dump (or back up) the contents of a database on the PostgreSQL system to a file
 - Script: SQL files
 - Archived: compressed binary files (using **pg_restore** to restore)

PostgreSQL Client Applications

- **pg_dumpall:** similar to the pg_dump program, except it dumps all of the databases to a file
- **pg_restore:** restore a PostgreSQL database from an archive file created by pg_dump

pgAdmin III / pgAdmin IV: a fancy graphical interface for administering a PostgreSQL system

– Open Source: www.pgadmin.org

